

# The ethics of disclosing the use of artificial intelligence tools in writing scholarly manuscripts

Research Ethics  
2023, Vol. 19(4) 449–465  
© The Author(s) 2023  
Article reuse guidelines:  
sagepub.com/journals-permissions  
DOI: 10.1177/17470161231180449  
journals.sagepub.com/home/rea  


**Mohammad Hosseini** 

Northwestern University Feinberg School of Medicine, USA

**David B Resnik** 

National Institute of Environmental Health Sciences, USA

**Kristi Holmes**

Northwestern University Feinberg School of Medicine, USA

## Abstract

In this article, we discuss ethical issues related to using and disclosing artificial intelligence (AI) tools, such as ChatGPT and other systems based on large language models (LLMs), to write or edit scholarly manuscripts. Some journals, such as *Science*, have banned the use of LLMs because of the ethical problems they raise concerning responsible authorship. We argue that this is not a reasonable response to the moral conundrums created by the use of LLMs because bans are unenforceable and would encourage undisclosed use of LLMs. Furthermore, LLMs can be useful in writing, reviewing and editing text, and promote equity in science. Others have argued that LLMs should be mentioned in the acknowledgments since they do not meet all the authorship criteria. We argue that naming LLMs as authors or mentioning them in the acknowledgments are both inappropriate forms of recognition because LLMs do not have free will and therefore cannot be held morally or legally responsible for what they do. Tools in general, and software in particular, are usually cited in-text, followed by being

## Corresponding author:

Mohammad Hosseini, Northwestern University Feinberg School of Medicine, 680 N Lake Shore Dr, Suite 1400, Chicago, IL 60611, USA.

Email: [mohammad.hosseini@northwestern.edu](mailto:mohammad.hosseini@northwestern.edu)



Creative Commons Non Commercial CC BY-NC: This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

mentioned in the references. We provide suggestions to improve APA Style for referencing ChatGPT to specifically indicate the contributor who used LLMs (because interactions are stored on personal user accounts), the used version and model (because the same version could use different language models and generate dissimilar responses, e.g., ChatGPT May 12 Version GPT3.5 or GPT4), and the time of usage (because LLMs evolve fast and generate dissimilar responses over time). We recommend that researchers who use LLMs: (1) disclose their use in the introduction or methods section to transparently describe details such as used prompts and note which parts of the text are affected, (2) use in-text citations and references (to recognize their used applications and improve findability and indexing), and (3) record and submit their relevant interactions with LLMs as supplementary material or appendices.

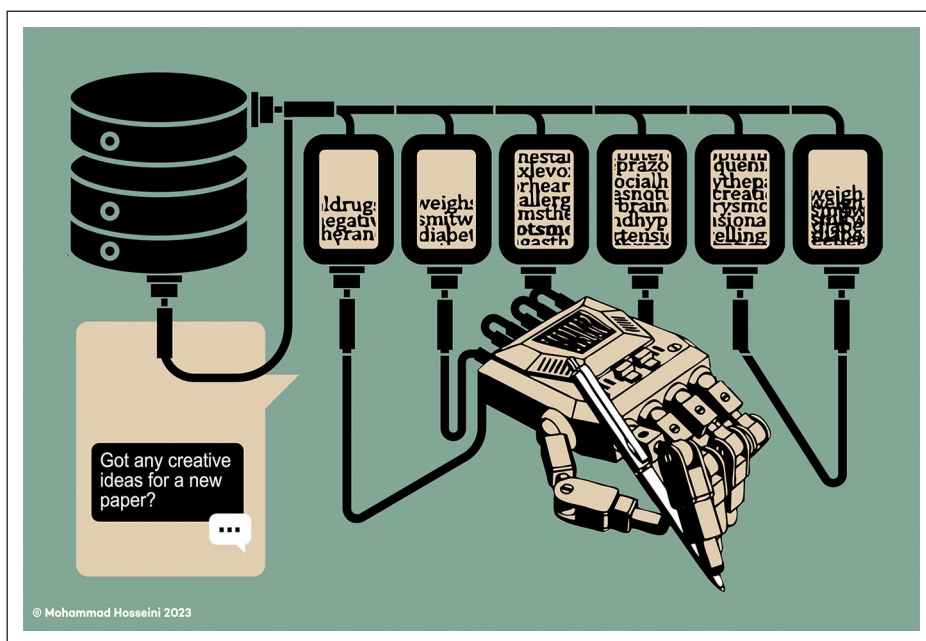
## Keywords

Publication ethics, authorship, transparency, large language models, ChatGPT, artificial intelligence, writing

OpenAI's ChatGPT and other systems based on large language models (LLMs), such as Elicit (Elicit, 2023) and Scholarcy (Scholarcy, 2023) are able to aggregate, summarize, paraphrase or write scholarly text. Some administrators at public schools, colleges, and universities have banned the use of artificial intelligence (AI) chatbots because they fear these technologies will undermine learning and academic integrity (Nolan, 2023). Many are predicting that LLMs will eliminate jobs that involve mid-level competence in computer programming and writing for media companies, advertisers, law firms, or other businesses (Cerullo, 2023). LLMs are also likely to transform scientific and scholarly research and communication in ways we cannot fully anticipate.

Articles and editorials published in journals, including *Nature* (Nature, 2023), *Accountability in Research* (Hosseini et al., 2023), *JAMA* (Flanagin et al., 2023), and *Science* (Thorp, 2023), as well as from the World Association of Medical Editors (Zielinski et al., 2023), have discussed the ethical issues raised by using LLMs, such as authorship, plagiarism, transparency, and accountability. While *Accountability in Research*, *JAMA*, and *Nature* decided to adopt or pursue policies that allow using LLMs under conditions that promote transparency, accountability, fair assignment of credit, and honesty, the editors of *Science* highlighted ethical problems created by LLMs and banned their use:

“... text written by ChatGPT is not acceptable: It is, after all, plagiarized from ChatGPT. Further, our authors certify that they themselves are accountable for the research in the paper. ... And an AI program cannot be an author. A violation of these policies will constitute scientific misconduct no different from altered images or plagiarism of existing works” (Thorp, 2023: 313).



**Figure 1.** ChatGPT and other LLMs have been and will be used by researchers.

There are three reasons for opposing journal policies that ban the use of LLMs in writing or editing scholarly manuscripts. First, bans are unenforceable. Even if prominent research institutions and journals were to adopt such measures, these efforts would likely be in vain, since detecting text that has been generated with LLMs is extremely difficult, partly because LLM-generated text can be altered by human beings to mask it. Although some companies, including OpenAI, have developed software designed to recognize LLM-generated text (Hu, 2023), these tools are unreliable and are likely to remain unreliable in finding LLM-generated text as computer scientists and researchers find ways of working around them. Second, bans may encourage undisclosed use of LLMs, which would undermine transparency and integrity in research and discourage training and education in responsible use of LLMs. Third, LLMs can play an important role in helping researchers who are not highly proficient in English (the *lingua franca* for most top journals) to write and edit their papers, or review others' manuscripts (Hosseini and Horbach, 2023), which could promote equity in science (Berdejo-Espinola and Amano, 2023).

As we will demonstrate in this article, LLMs such as ChatGPT have been, and will be used by researchers in various ways (Figure 1). Ethical principles, including openness, honesty, transparency, efficient use of resources, and fair allocation of credit (Shamoo and Resnik, 2022) demand disclosing the use of LLMs. Openness, transparency, and honesty about used methods and tools are paramount to fostering integrity, reproducibility, and rigor in research. To the extent that

**Table 1.** Evaluation of different policy options concerning the use AI in writing or editing scholarly publications.

| Policy Option                                                     | Rationale                                                                                                                                                   | Problems                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ban the use of AI in generating texts for scholarly manuscripts   | <ul style="list-style-type: none"><li>• Avoids difficult issues related to fair allocation of authorship credit, accountability, and transparency</li></ul> | <ul style="list-style-type: none"><li>• Not enforceable</li><li>• Leads to clandestine use of AIs</li><li>• Discourages equity in science and prevents helping researchers who are not adept at writing in languages other than their first language</li></ul> |
| Allow AIs to be listed as authors                                 | <ul style="list-style-type: none"><li>• Avoids giving human authors undue credit for work done by AIs</li><li>• Promotes transparency</li></ul>             | <ul style="list-style-type: none"><li>• AIs cannot be morally or legally responsible or accountable</li></ul>                                                                                                                                                  |
| Allow AIs to be listed in the acknowledgments section             | <ul style="list-style-type: none"><li>• Promotes transparency</li></ul>                                                                                     | <ul style="list-style-type: none"><li>• AIs cannot be morally or legally responsible or accountable</li></ul>                                                                                                                                                  |
| Disclose use of AIs in the body of the texts and among references | <ul style="list-style-type: none"><li>• Promotes transparency</li><li>• Consistent with disclosing the use of other tools</li></ul>                         | <ul style="list-style-type: none"><li>• Consistency of disclosure</li></ul>                                                                                                                                                                                    |

disclosure facilitates replicating a completed study or support for future studies, it also promotes efficient uses of resources. With respect to fair allocation of credit, not disclosing LLMs, especially those which provide context-specific suggestions and can generate or substantially affect content, violates norms of ethical attribution because it results in giving undue credit to (human) contributors for work which they did not do (Verhoeven et al., 2023).

We believe that a concerted effort will be required to use LLMs responsibly and promote ethical and transparent disclosure in scholarly work. Toward this end, we will argue that LLMs should not be named as authors or mentioned in the acknowledgments because they do not have free will and therefore cannot be held morally or legally responsible for what they do. Use of LLMs, like the use of other type of software tools, should be cited in-text, followed by being mentioned in the references. (See Table 1 for assessment of different policy options for use of LLMs in scholarly publishing.)

**LLMs as authors?**

In a paper titled “AI-assisted authorship: How to assign credit in synthetic scholarship,” Jenkins and Lin (2023) argue that LLMs should be named as authors if they make substantial contributions to publications (and other products, such as art-work) that would be worthy of credit if they were done by human beings. Without

question, LLMs can make substantial contributions that are not readily distinguishable from the contributions made by human beings. Although LLMs can make some glaring mistakes, are susceptible to bias, and may even fabricate facts or citations (Hosseini et al., 2023), these flaws should not be held against them because human researchers might make similar errors. According to Jenkins and Lin, when LLMs make substantial contributions that are on par with human contributions, they should be credited as such. Failing to do so would assign credit inappropriately to human authors (Verhoeven et al., 2023).

Some researchers have already embraced this idea by naming LLMs as authors. For example, in an editorial titled “Open artificial intelligence platforms in nursing education: Tools for academic progress or abuse?” published in the journal of *Nurse Education in Practice*, ChatGPT is listed as the second author (O’Connor, 2023). O’Connor notes that the first five paragraphs of this piece were written by ChatGPT in response to provided prompts. Another example of listing an LLM as an author is a paper titled “Rapamycin in the context of Pascal’s Wager: generative pre-trained transformer perspective,” published in the journal of *Oncoscience* (Zhavoronkov, 2022).

Although it is important to disclose how an LLM has been used to write or edit a manuscript, designating an LLM as an author is ethically problematic because widely accepted journal guidelines, such as those provided by the International Committee of Medical Journal Editors (ICMJE), and research norms, such as those articulated by Shamoo and Resnik (2022) and Briggles and Mitcham (2012), imply that authors must be willing to be responsible and accountable for the content of the manuscript. Accountability and credit are two sides of the same coin, and contributors cannot have one without the other (Hosseini et al., 2022; Resnik, 1997; Smith, 2017).

Accountability and responsibility are closely related, but different concepts (Davis, 1995). Today’s LLMs are neither responsible nor accountable because they lack free will (or self-determination). To be accountable for an action, one must be able to explain it to others and be subject to its legal and moral consequences, which implies responsibility. For example, if a driver crashes their car into a pottery store, the legal system could hold them accountable in various ways: they may need to pay for caused damages, pay a fine, or explain their conduct to a judge or jury, and they may even lose their driver’s license. However, the legal system would not hold a young child accountable for breaking a plate in a pottery shop because the child is not responsible for their actions. The legal system might, however, hold the child’s parents responsible for not supervising the child more closely and also hold them to account by requiring them to pay for the damage.

One can be held morally and legally responsible for an action only if that action results from one’s free choices (Mele, 2006). There is a long-standing philosophical debate about whether human beings have free will and what free will amounts to, which we do not need to engage here. The sense of “free” we have in mind need

not be metaphysically robust but should capture the sense of the word used in ethics, law, and ordinary language (Manson and O'Neill, 2007; Mele, 2006). An action is free (i.e. self-determined) in this metaphysically limited sense if it results from the individual's deliberate choices. For an individual to make a deliberate choice, they must have consciousness, self-awareness, understanding, the ability to reason, information, and values or preferences (see Mele, 2006; O'Connor, 2022). Current LLMs do not have the capacities needed to make free choices. While they can manipulate linguistic symbols and digital data quite adeptly, they lack consciousness, self-awareness, a humanlike understanding of language, and values or preferences (Bogost, 2022; Teng, 2020). AIs may have these capacities in the future, but that remains to be seen.

In summary, LLMs should not be named as authors because they cannot be held legally and morally responsible for what they do, and authorship implies responsibility (Copyright Review Board, 2022; Shamoo and Resnik, 2022). The view defended here is also expressed in a recent position statement published by the Committee on Publication Ethics (COPE):

“AI tools cannot meet the requirements for authorship as they cannot take responsibility for the submitted work. As non-legal entities, they cannot assert the presence or absence of conflicts of interest nor manage copyright and license agreements” (COPE Position Statement, 2023: para. 2).

In research, accountability is essential for promoting integrity, reproducibility, rigor, and other important epistemic and moral values (Shamoo and Resnik, 2022). Because LLMs cannot be held morally or legally responsible, they also cannot be held accountable for their actions. If there are questions about the validity of the data or methods in a paper published in a scientific journal, the authors must be able to explain what they did and why, and be prepared to take appropriate steps to address errors or ethical transgressions, such as submitting a correction or retraction to the journal.

Exploring the two mentioned examples wherein LLMs were named authors would help further clarify our stance. Concerning the S. O'Connor editorial mentioned earlier, while taking responsibility for the content is not a requirement for authorship according to *Nurse Education in Practice* guidelines (thus, making irrelevant the argument that LLMs cannot be held accountable), the editorial does not meet the journal's own authorship guidelines for another reason.<sup>1</sup> *Nurse Education in Practice* authorship guidelines note:

“All authors should have made substantial contributions to all the following: (1) the conception and design of the study, or acquisition of data, or analysis and interpretation of data; (2) drafting the article or revising. You will be asked to confirm this on submission critically for important intellectual content; and (3) final approval of the version to be submitted. Everyone who meets these criteria should be listed as an author.” (Nurse Education in Practice, 2023: para. 19)



We understand from the editorial that ChatGPT drafted five (out of seven) paragraphs, thus meeting the first two criteria. However, it did not approve the final version of the manuscript because approval is a form of consent and one cannot consent to something without free will, which ChatGPT and other LLMs do not currently have, despite some sensational claims to the contrary (de Cosmo, 2022).

Regarding the Zhavoronkov article, *Oncoscience*'s guidelines also require that authors give final approval:

“As a general guideline, persons listed as authors should have contributed substantively to (1) the conception and design of the study, acquisition of data, or analysis and interpretation of data; (2) drafting of the article or revising it for important content; and 3) final approval of the version to be published” (Oncoscience, 2023: para. 23).

To get around this problem, Zhavoronkov claims to have received final approval from Sam Altman, the co-founder and Chief Executive Officer (CEO) of OpenAI, which owns and operates ChatGPT:

“[D]ue the fact that the majority of the article was produced by the large language model, to set a precedent, the decision was made to include ChatGPT as a co-author and add the appropriate explanation and reference in the article. ChatGPT also assisted with references and appropriate formatting. Alex Zhavoronkov reached out to Sam Altman, the co-founder and CEO of OpenAI to confirm, and received a response with no objections” (Zhavoronkov, 2022: 84).

However, approval from the CEO of OpenAI should not be considered approval from the author, any more than approval by the corresponding author of a paper should count as approval by other (human) authors. In theory, an author could designate another party to grant approval for them, but doing so would also require consent, which, as we have already argued, LLMs cannot give. We also note that *Oncoscience*'s guidelines apply to “persons listed as authors”<sup>2</sup> and, for the reasons discussed above, LLMs are not persons, hence, this authorship designation does not meet journal's own authorship criteria.

Jenkins and Lin (2023) object to the arguments that AIs cannot be named as authors because they lack accountability and because they cannot approve the final version by pointing out that authorship is sometimes granted posthumously, even though people who are dead cannot be held accountable or approve anything:

“*Nature* also argues AI writers should not be credited as authors on the grounds that they cannot be accountable for what they write. This line of argument needs to be considered more carefully. For instance, authors are sometimes posthumously credited, even though they cannot presently be held accountable for what they said when alive, nor can they approve of a posthumous submission of a manuscript; yet it would clearly be hasty to forbid the submission or publication of posthumous works” (Jenkins and Lin, 2023: 3).

However, we believe the ethical acceptability of posthumous authorship is not a convincing objection to our view because posthumous authors were capable of being held accountable and of approving the final version of the paper when they did the work they are credited for. A posthumous author is someone who would have been able to take responsibility and would have approved the final version, if they were alive. While it is possible that this sometimes might not be the case (e.g. a work might be irresponsibly or even maliciously attributed to someone who would have not agreed to be an author if they were alive), generally posthumous authorship is granted for works that individuals contributed to, and would endorse if they were alive.

Moreover, posthumous authorship is a way of valuing and affirming a person's contributions to research and their status and reputation. Authorship, in this sense, is a form of social capital that is awarded, protected, and exchanged in human relationships (Smith, 2017). These relationships are very important when a person is alive and actively pursuing their interests and goals and persist after death. Indeed, the legal system recognizes rights that continue after a person has died, such as rights concerning the disposition of one's estate, rights to privacy, and intellectual property rights (Mennell and Burr, 2017; Miller and Davis, 2018). Since LLMs are not persons, these social aspects of authorship do not apply to them, but they do apply to human authors, even dead ones.

Polonsky and Rotman (2023) object to the requirement that AI tools should be denied authorship because they are not human beings by pointing out that authorship credit is sometimes granted to groups, such as corporations, government entities, and research centers. However, this objection is misleading because corporations, government entities, and research centers can own copyrights,<sup>3</sup> can be held morally and legally responsible and accountable and can even approve the final version of a manuscript. Responsibility here is at the level of the group as opposed to the individual but it is responsibility, nonetheless. We note that International Committee of Medical Journal Editors [ICMJE] (2023: p. 3) guidelines allow for groups to be named as authors if they can take responsibility: "Some large multi-author groups designate authorship by a group name, with or without the names of individuals. When submitting a manuscript authored by a group, the corresponding author should specify the group name if one exists, and clearly identify the group members who can take credit and responsibility for the work as authors."

A future scenario considered by Lee (2023) is when LLMs develop to the point where they can explain to a human being what they have done and why. The explainable AI movement seeks to make this type of interaction possible (Ankarstad, 2020). Lee (2023) argues that LLMs should be credited with authorship if the day arrives when they can clearly explain what they have written and why, but we disagree. Although being able to explain what they have done and



why would take LLMs a step closer to being accountable, it would still fall far short of the degree of accountability we expect from human beings. Part of being accountable is not only being able to explain one's conduct but being able to face the consequences of it, such as punishment. Researchers who fabricate or falsify data can be subject to various forms of punishment, such as loss of funding or employment, reputational damage, and, in rare cases, imprisonment (Shamoo and Resnik, 2022). These and other forms of punishment play an important role in deterring misconduct in research (Horner and Minifie, 2011), but punishments cannot affect (let alone deter) LLMs in any way, because they do not have interests, values, or feelings. While it might be true that some sanctions, such as banning the use of a specific application in certain research contexts or financial penalties, could impact investors or developers and encourage them to develop better applications, these would not constitute punishment for LLMs, which may have provided biased analyses or made mistakes that resulted in ethical catastrophes.

Nothing mentioned in this section should be taken to imply that from an ethical perspective, AIs can never be authors of scholarly work. If AIs develop to the point where there is compelling evidence that they have free will and can be held responsible and accountable and can participate in society like humans, then they could be named as authors on scholarly publications. As we said earlier that day has not yet come, but it may be approaching faster than many people think.

## Recognizing LLMs in the acknowledgments section

If LLMs cannot be co-authors, should they be mentioned in the acknowledgments section? After all, non-author contributors are typically recognized there. Recognizing non-authors in the acknowledgments section is also supported by widely accepted guidelines such as those provided by the ICMJE (2023: p. 3): “Those whose contributions do not justify authorship may be acknowledged individually or together as a group . . . Because acknowledgment may imply endorsement by acknowledged individuals of a study's data and conclusions, editors are advised to require that the corresponding author obtain written permission to be acknowledged from all acknowledged individuals.”

This approach is endorsed by some. For example, Jenkins and Lin (2023) and Hughes-Castleberry (2023) argue that LLMs could be named in the acknowledgments section. A *Nature* news article quoted Magdalena Skipper, editor-in-chief of *Nature* in London that those using LLMs in any way while developing a paper “should document their use in the methods or *acknowledgments sections*” [emphasis added] (Stokel-Walker, 2023: 620). Sabina Alam, the director of publishing ethics and integrity at Taylor & Francis has defended the same position, “authors are responsible for the validity and integrity of their work, and should cite any use

of LLMs *in the acknowledgments section*” [emphasis added] (Stokel-Walker, 2023: 620).

We believe that crediting LLMs in the acknowledgments section of a manuscript is inappropriate for largely the same reasons that LLMs should not be named as authors, that is, because they lack free will and therefore cannot consent to being acknowledged. Although a mention in the acknowledgments section of a paper is not as prestigious as an author byline, it still carries some moral and legal weight and should therefore involve consent. If a person mentioned in the acknowledgments section provided data for a study that comes under suspicion of fabrication or falsification, they may be held morally (and perhaps legally) responsible and accountable for the integrity of the data they provided. Moreover, a person may not want to be mentioned in the acknowledgments section if they disagree with the conclusions of a study and do not want to be associated with it.

If LLMs are merely a tool, recognizing their use should be consistent with how other tools are recognized. For example, researchers use search engines, such as Google, and scholarly indices, such as PubMed and Web of Science, to search the extant literature and find resources or use software like SPSS to identify relationships and correlations, but none of these tools are mentioned in the acknowledgments section. Why should LLMs be mentioned? One might argue that LLMs such as ChatGPT, not only search through available literature and analyze data but also can verbalize their observations and prepare manuscripts (see e.g. Jabotinsky and Sarel, 2022; Polonsky and Rotman, 2023), and so their contribution is comparable to that of a human. However, as mentioned earlier, recognition is not only about assigning credit but also involves responsibility and accountability, and LLMs cannot be considered responsible or accountable in the way human beings can be.

## **Disclosing the use of LLMs in the body of the text**

Tools used in research are typically disclosed in the body of the text, and in the case of software applications, they are also cited with in-text citations and in the references (Katz et al., 2021). Given their capabilities and complexities, how should LLMs and their use be described in the body of the text?

An example how this can be done appeared in a recent preprint in which the authors describe their use of ChatGPT (Blanco-Gonzalez et al., 2022). However, disclosure by mentioning this information only in the text also presents challenges, especially in terms of findability of articles that used LLMs, due to issues such as a lack of indexing (in the case of non-English content) and access to the full article text (in the case of paywalled content), or consistency of disclosures which could impact openness and transparency (e.g. if some studies under report the use of LLMs). In Blanco-Gonzalez et al. (2022: p.2), the authors highlighted the extent

of their use (i.e. “total percentage of similarity between the preliminary text, obtained directly from ChatGPT, and the current version of the manuscript”) and added that 33.9% of the manuscript comprise of text generated by ChatGPT and is used verbatim or after revision (“identical 4.3%, minor changes 13.3% and related meaning 16.3%”). This level of detail is unlikely to be provided consistently by all researchers and is perhaps impossible to calculate when LLMs contribute to tasks that are not quantifiable, such as conceptualization. More importantly, this information still does not let readers know which part of the text has been written by LLMs.

Both challenges (i.e. findability of articles that used LLMs and identifying what part of the text is affected by their use) could be resolved via general norms of software citation that include in-text citations and referencing. In fact, APA style has already provided guidelines about in-text citations and referencing ChatGPT (McAdoo, 2023) and notes that disclosure could be different depending on the article type. APA advises disclosure in the methods section in research articles or in the introduction in literature reviews, essays or response or reaction papers (McAdoo, 2023).

Indeed, in-text citations offer the required signposting to indicate what part of the text is affected by LLMs. In manuscripts behind a paywall, citations are not accessible to all readers, but corresponding references are often open, and thanks to open citations initiatives (e.g. I4OC) will likely become more accessible. That said, ensuring the consistency of disclosures could be challenging (similar challenges are faced in software citation, e.g. see Li et al., 2017) and could be addressed through training and education, as well as promoting best practices.

The template offered by the APA style (McAdoo, 2023: para. 5) recommends the following format for description of use and in-text citation and referencing:

“When prompted with “Is the left brain right brain divide real or a metaphor?” the ChatGPT-generated text indicated that although the two brain hemispheres are somewhat specialized, “the notation that people can be characterized as ‘left-brained’ or ‘right-brained’ is considered to be an oversimplification and a popular myth” (OpenAI, 2023).

## Reference

OpenAI (2023). ChatGPT (Mar 14 version) [Large language model]. <https://chat.openai.com/chat>”

We suggest slight modifications to the suggested referencing style, to ensure that responsibilities and accountabilities are distributed fairly, and use cases are disclosed more transparently. Let’s not forget that LLMs learn fast and change rapidly, making it vital to disclose not only what version was used but also *which model, when* and by *who*.

*Which model?* As per May 2023, when using ChatGPT PLUS (the paid version), one can choose between two different models (GPT-3.5 and GPT-4) from the same version (ChatGPT May 12 Version) to generate text. According to the developers, each of these versions offers different degrees of reasoning, speed, and conciseness, but more importantly, they provide dissimilar responses to the same prompt.

*When?* Since LLMs are constantly learning (or in the event of plugging them to the internet, receive new data), responses to the same question few days or weeks apart could be different as was shown recently (Hosseini and Horbach, 2023).

*By who?* An indication of who used the system would be vital to better delineate responsibilities. Especially in systems like ChatGPT that can generate dissimilar responses to similar prompts, and also store previous interactions on individual user accounts, collecting this information is required to ensure openness and transparency.

On that basis, when mentioning LLMs among references, it would be necessary to include information about the used version, the used model, the date of use as well as the user's name. Accordingly, we suggest the following referencing format:

OpenAI (2023). ChatGPT (GPT-4, May 12 Version) [Large language model]. Response to query made by X.Y. Month/Day/Year. <https://chat.openai.com/chat>

### ***Best practices for disclosure of using LLMs***

Given the considerations raised about disclosure as co-authors or in the acknowledgments section, we recommend that scholarly community disclose their use of LLMs using other means. Our suggestions combine insights offered by the journal of *Accountability in Research* and APA style guidelines.

To uphold ethical norms of transparency, openness, honesty, and fair attribution of credit, in cases where LLMs are used, disclosure should happen:

- (1) As free text in the introduction or methods section (to honestly and transparently describe details about who used LLMs, when, how, using what prompts and disclose what sections of the text are affected; to prevent giving undue credit to human contributors for work they did not do)
- (2) Through in-text citations and among references (to improve findability and indexing) using the following format:

OpenAI (2023). ChatGPT (GPT-4, May 12 Version) [Large language model]. Response to query made by X.Y. Month/Day/Year. <https://chat.openai.com/chat>

To enable verification, interactions with LLMs (including specific prompts, and dates of query) should be recorded and disclosed:

(3) As supplementary material or in appendices

Clearly, since LLMs may be used differently in various research areas or in different research outputs, more detailed guidelines or specific requirements about the use of LLMs could be developed by professional associations or journal editors. An example of such effort was demonstrated by organizers of the 40th International Conference on Machine Learning (ICML) who noted among conference policies “Papers that include text generated from a large-scale language model (LLM) such as ChatGPT are prohibited unless these produced text is presented as a part of the paper’s experimental analysis (ICML 2023: para 8).”

One may ask whether the use of an LLM should be disclosed if it is used only in ways that do not generate or substantially affect content, such as to improve grammar, correct typos, or provide suggestions for alternative words or phrases, like Grammarly, or other writing-assistance programs already do. While we think it is not necessary to disclose the use of LLMs if they are only used in ways that do not generate or substantially affect content, we think that this situation will be rare because LLMs can do so much more than correct grammatical or typographical errors. When LLMs are used to edit and rewrite manuscripts, they are likely to generate or substantially affect content. Thus, we think the best practice will still be to disclose the LLMs in writing or editing.

One might also ask whether LLM use should be disclosed if they are incorporated into existing word processing programs, such as MS Word, which is likely to happen soon (Kelly, 2023). Our answer, again, would be that LLM use should be disclosed if the LLM generates or substantially affects content. If this use happens as part of a word processing program, then that should be mentioned in the disclosure.

## Conclusion

The use of LLMs, such as ChatGPT, to write, review and edit scholarly manuscripts presents challenging ethical issues for researchers and journals. We argue that banning the use of LLMs would be a mistake because a ban would not be enforceable and would encourage undisclosed use of LLMs. Also, since LLMs can have some useful applications in writing and editing text (especially for those conducting research in a language other than their first language), banning them would not support diversity and inclusion in scholarship. The most reasonable response

to the dilemmas posed by LLMs is to develop policies that promote transparency, accountability, fair allocation of credit, and integrity. The use of LLMs should be disclosed through (1) free-text in the introduction or methods section, (2) in-text citations and references, (3) supplementary materials or appendices. LLMs should not be named as authors or credited in the acknowledgments section because they lack free will and cannot be held morally or legally responsible.

### **Acknowledgements**

We thank the journal editor and four anonymous reviewers for their constructive and valuable feedback. We are grateful for helpful comments from Lisa Rasmussen and Daniel Carey.

### **Author's contributions**

M.H. Conceptualization, Investigation, Project Administration, Writing-Original Draft, Writing-Review & Editing.

D.B.R Conceptualization, Investigation, Supervision, Writing-Original Draft, Writing-Review & Editing.

K.H. Funding acquisition, Writing-Review & Editing.

### **Declaration of conflicting interest**

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

### **Funding**

All articles in Research Ethics are published as open access. There are no submission charges and no Article Processing Charges as these are fully funded by institutions through Knowledge Unlatched, resulting in no direct charge to authors. For more information about Knowledge Unlatched please see here: <http://www.knowledgeunlatched.org> This research was supported by the National Institutes of Health (NIH) through the Intramural Program of the National Institute of Environmental Health (NIEHS) and the National Center for Advancing Translational Sciences (NCATS, UL1TR001422). The funders have not played a role in the design, analysis, decision to publish, or preparation of the manuscript. This work does not represent the views of the NIEHS, NCATS, NIH, or US government.

### **Data and materials availability**

Not applicable.

### **ORCID iDs**

Mohammad Hosseini  <https://orcid.org/0000-0002-2385-985X>

David B. Resnik  <https://orcid.org/0000-0002-5139-9555>

### **Notes**

1. It is important to note that S. O'Connor (2023) published a corrigendum to this editorial that removed ChatGPT as an author.



2. Oncoscience authorship guidelines read “As a general guideline, persons listed as authors should have contributed substantively to (1) the conception and design of the study, acquisition of data, or analysis and interpretation of data; (2) drafting of the article or revising it for important content; and (3) final approval of the version to be published.” (Oncoscience, 2023).
3. Group authorship grant copyrights for the group (or institution) because they refer to “work made for hire,” that is, work that is within the scope of one’s employment agreement (Lee, 2023: 3). As mentioned earlier, machines cannot be copyright holders.

## References

- Ankarstad A (2020) What is explainable AI (XAI)? Available at: <https://towardsdatascience.com/what-is-explainable-ai-xai-afc56938d513> (accessed 10 April 2023).
- Berdejo-Espinola V and Amano T (2023) AI tools can improve equity in science. *Science* 379(6636): 991.
- Blanco-Gonzalez A, Cabezon A, Seco-Gonzalez A, et al. (2022) The Role of AI in Drug Discovery: Challenges, Opportunities, and Strategies. arXiv:2212.08104. [Computation and Language]. [arXiv]
- Bogost I (2022) ChatGPT is dumber than you think. *The Atlantic*. Available at: <https://www.theatlantic.com/technology/archive/2022/12/chatgpt-openai-artificial-intelligence-writing-ethics/672386/> (accessed 7 December 2022)
- Briggle A and Mitcham C (2012) *Ethics and Science: An Introduction*. Cambridge: Cambridge University Press.
- Cerullo M (2023) These jobs are most likely to be replaced by chatbots like ChatGPT. *CBS News*. Available at: <https://www.cbsnews.com/news/chatgpt-artificial-intelligence-chatbot-jobs-most-likely-to-be-replaced/> (accessed 1 February 2023).
- COPE Position Statement (2023). Available at: <https://publicationethics.org/cope-position-statements/ai-author> (accessed 15 February 2023)
- Copyright Review Board (2022) Re: Second Request for Reconsideration for Refusal to Register A Recent Entrance to Paradise (Correspondence ID 1-3ZPC6C3; SR # 1-7100387071). Available at: <https://www.copyright.gov/rulings-filings/review-board/docs/a-recent-entrance-to-paradise.pdf> (accessed 10 April 2023)
- Davis M (1995) A preface to accountability in the professions. *Accountability in Research* 4(2): 81–90.
- de Cosmo L (2022) Google engineer claims AI chatbot is sentient: Why that matters. *Scientific American*. July 12, 2022. Available at: <https://www.scientificamerican.com/article/google-engineer-claims-ai-chatbot-is-sentient-why-that-matters/> (accessed 1 April 2023)
- Elicit (2023). Available at: <https://elicit.org/> (accessed 10 April 2023)
- Flanagin A, Bibbins-Domingo K, Berkwits M, et al. (2023) Nonhuman “Authors” and implications for the integrity of scientific publication and Medical Knowledge. *JAMA* 329: 637.
- Horner J and Minifie FD (2011) Research Ethics III: Publication Practices and authorship, conflicts of interest, and research misconduct. *Journal of Speech Language and Hearing Research* 54(1): S346–S362.
- Hosseini M and Horbach SPJM (2023) Fighting reviewer fatigue or amplifying bias? *Considerations and recommendations for use of ChatGPT and other Large Language Models in scholarly peer review*. Research Integrity and Peer Review. 8(1):4.
- Hosseini M, Lewis J, Zwart H, et al. (2022) An ethical exploration of increased average number of authors per publication. *Science and Engineering Ethics* 28(3): 25.

- Hosseini M, Rasmussen LM and Resnik DB (2023) Using AI to write scholarly publications. *Accountability in Research* 0(0): 1–9.
- Hughes-Castleberry K (2023) From Cats to Chatbots: How Non-Humans Are Authoring Scientific Papers. *Discover Magazine*. Available at: <https://www.discovermagazine.com/the-sciences/from-cats-to-chatbots-how-non-humans-are-authoring-scientific-papers> (accessed 7 April 2023)
- Hu K (2023) ChatGPT owner launches 'imperfect' tool to detect AI-generated text. Reuters. Available at: <https://www.reuters.com/business/chatgpt-owner-launches-imperfect-tool-detect-ai-generated-text-2023-01-31/> (accessed 1 February 2023)
- ICML 2023 (n.d) International Conference on Machine Learning - ICML. Available at: <https://icml.cc/Conferences/2023/CallForPapers> (accessed 12 April 2023)
- International Committee of Medical Journal Editors (2023) Defining the Role of Authors and Contributors. <https://www.icmje.org/recommendations/browse/manuscript-preparation/preparing-for-submission.html> (accessed 10 April 2023)
- Jabotinsky HY and Sarel R (2022) *Co-authoring with an AI? Ethical Dilemmas and Artificial Intelligence* (SSRN Scholarly Paper No. 4303959). <https://doi.org/10.2139/ssrn.4303959>
- Jenkins R and Lin P (2023) *AI-Assisted Authorship: How to Assign Credit in Synthetic Scholarship* (SSRN Scholarly Paper No. 4342909). <https://doi.org/10.2139/ssrn.4342909>
- Katz DS, Hong NPC, Clark T, et al. (2021) *Recognizing the value of software: A software citation guide* (9:1257). F1000Research. <https://doi.org/10.12688/f1000research.26932.2>
- Kelly SM (2023) *Microsoft is Bringing ChatGPT Technology to Word, Excel and Outlook*. CNN. Available at: <https://www.cnn.com/2023/03/16/tech/openai-gpt-microsoft-365/index.html> (accessed 16 March 2023).
- Lee JY (2023) Can an artificial intelligence chatbot be the author of a scholarly article? *Journal of Educational Evaluation for Health Professions* 20: 6.
- Li K, Yan E and Feng Y (2017) How is R cited in research outputs? Structure, impacts, and citation standard. *Journal of Informetrics* 11(4): 989–1002.
- Manson NC and O'Neill O (2007) *Rethinking Informed Consent in Bioethics*. Cambridge: Cambridge University Press.
- McAdoo T (2023) How to cite ChatGPT. *APA Style Blog*. Available at: <https://apastyle.apa.org/blog/how-to-cite-chatgpt> (accessed 17 May 2023)
- Mele A (2006) *Free Will and Luck*. Oxford: Oxford University Press.
- Mennell R and Burr S (2017) *Wills and Trusts*. St. Paul, MN: West Publishing.
- Miller A and Davis M (2018) *Intellectual Property*. St. Paul, MN: West Publishing.
- Nature (2023) Tools such as ChatGPT threaten transparent science; here are our ground rules for their use. *Nature* 613(7945): 612–612.
- Nolan B (2023) Here are the schools and colleges that have banned the use of ChatGPT over plagiarism and misinformation fears. *Business Insider*, January 30, 2023. Available at: <https://www.businessinsider.com/chatgpt-schools-colleges-ban-plagiarism-misinformation-education-2023-1> (accessed 1 April 2023)
- Nurse Education in Practice (2023) Guide for authors. Available at: <https://www.elsevier.com/journals/nurse-education-in-practice/1471-5953/guide-for-authors> (accessed 1 April 2023)
- Oncoscience (2023) Editorial policies. Available at: <https://www.oncoscience.us/editorial-policies/> (accessed 1 April 2023)
- O'Connor C (2022) Free will. *Stanford Encyclopedia of Philosophy*. Available at: <https://plato.stanford.edu/entries/freewill/>

- O'Connor S (2023) Corrigendum to "Open artificial intelligence platforms in nursing education: Tools for academic progress or abuse?" [Nurse Educ. Pract. 66 (2023) 103537]. *Nurse Education in Practice* 67: 103572.
- O'Connor S; ChatGPT (2023) Open artificial intelligence platforms in nursing education: Tools for academic progress or abuse? *Nurse Education in Practice* 66: 103537.
- Polonsky M and Rotman J (2023) *Should Artificial Intelligent (AI) Agents be Your Co-author? Arguments in favour, informed by ChatGPT* (SSRN Scholarly Paper No. 4349524). <https://doi.org/10.2139/ssrn.4349524>
- Resnik DB (1997) A proposal for a new system of credit allocation in science. *Science and Engineering Ethics* 3: 237–243.
- Scholarcy (2023). Available at: <https://www.scholarcy.com/> (accessed 1 April 2023)
- Shamoo AE and Resnik DB (2022) *Responsible Conduct of Research*, 4th edn. New York, NY: Oxford University Press.
- Smith E (2017) A theoretical foundation for the ethical distribution of authorship in multidisciplinary publications. *Kennedy Institute of Ethics Journal* 27(3): 371–411.
- Stokel-Walker C (2023) ChatGPT listed as author on research papers: many scientists disapprove. *Nature* 613(7945): 620–621.
- Teng CH (2020) Free will and AI. *Becoming Human*. Available at: <https://becominghuman.ai/free-will-and-ai-85adbb09ac07> (accessed 20 February 2020)
- Thorp HH (2023) ChatGPT is fun, but not an author. *Science* 379(6630): 313–313.
- Verhoeven F, Wendling D and Prati C (2023) ChatGPT: When artificial intelligence replaces the rheumatologist in medical writing. *Annals of the Rheumatic Diseases*. Epub ahead of print 11 April 2023. DOI: 10.1136/ard-2023-223936.
- ChatGPT Generative pre-trained transformer perspective, Zhavoronkov A (2022) Rapamycin in the context of Pascal's Wager: Generative pre-trained transformer perspective. *Oncoscience* 9: 82–84.
- Zielinski C, Winker M, Aggarwal R, et al. (2023) Chatbots, ChatGPT, and Scholarly Manuscripts. *WAME Recommendations on ChatGPT and Chatbots in Relation to Scholarly Publications*. Available at: <https://wame.org/page3.php?id=106> (accessed 10 April 2023)